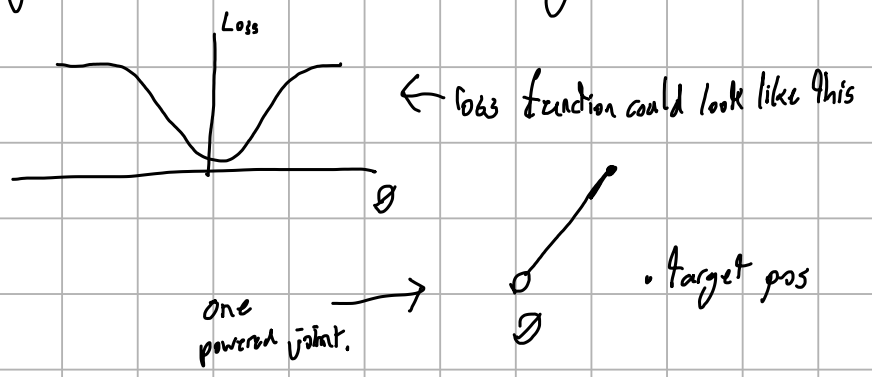
1) a) no in quaternion it is scaled by $\sin(\frac{\theta}{2})$.

b) ZZZ only rotates on the Z axis, and doesn't cover all possible rotations in 3d. Quaternion doesn't have gimbal lock, but is hard to visualize. ZXZ has gimbal lock but is easy to visualize.

c) Not always, it depends how complicated your system. You could have a very simple arm with a simple loss scape with just one local minimum = global minimum




d) $w = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} q_0 \\ q_1 \\ q_2 \\ q_3 \end{bmatrix} = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ w_0 \sin(\frac{\theta}{2}) \\ w_1 \sin(\frac{\theta}{2}) \\ w_2 \sin(\frac{\theta}{2}) \end{bmatrix} = \boxed{\begin{bmatrix} \frac{\sqrt{2}}{2} \\ 0 \\ 0 \\ \frac{\sqrt{2}}{2} \end{bmatrix}}$$